

Designing for Qualitative Evaluation of Synthetic Medical Data

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Abstract

Machine learning in healthcare often struggles with data access for model training due to privacy restrictions, rare conditions, and high acquisition costs. Synthetic data offers a potential workaround, yet there are no agreed-upon gold standards for evaluating it. As quantitative metrics alone cannot fully assess the desired qualities of generative model outputs, human inspection is a key component of validation, warranting a “Doctor-in-the-loop” approach. However, research is scarce on best practices for interaction and user interface design in such systems. This paper presents preliminary designs for qualitative synthetic medical data evaluation, informed by four participatory workshops with seven doctors and nine machine learning engineers. Spanning tabular, image, and time series data, this study emphasised transparency and clear communication of the synthetic data generation. In addition to presenting the rationale behind the evaluation workflow design, we highlight challenges in the medical domain, including doctors’ limited familiarity and skepticism with synthetic data.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**; **User studies**; **Human computer interaction (HCI)**;

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Keywords

Synthetic Data (SD), Synthetic medical data (SMD), Doctor-in-the-Loop (DITL), Qualitative Evaluation, Participatory Design, Machine Learning in Healthcare, Human-Computer Interaction (HCI)

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1 Introduction

Synthetic data (SD) generation has surged in popularity due to its promise to provide large, high-quality datasets needed for machine learning (ML) training, particularly in healthcare. In this context access to data is often limited due to patient privacy, data scarcity on rare conditions, and steep costs for acquiring and annotating new data [10, 22]. SD —artificial data generated through a specialised mathematical model or algorithm explicitly intended to address one or more data science tasks [15]— is already being used in healthcare for different purposes, including simulation and prediction (e.g., early diagnosis of cerebellar ataxia [16]); healthcare IT development (e.g., testing and demonstration of software applications), and education and training (e.g., teaching data science techniques using synthetic cardiovascular data [10]).

In addition to its applications in healthcare, SD is also being explored in other areas to augment datasets and enhance the performance of ML models. For example, it has been applied to a public news dataset to improve the effectiveness of a fake news detection ML model [24]. Similarly, researchers are examining its potential to enhance ML models for object detection in computer vision [25].

However, there are concerns about SD in healthcare—the risk of data leakage, low transparency [8], overgeneralization, and the misrepresentation of demographic and medical profiles due to incorrect statistical correlations [9]. Additionally, there is a lack of consensus on methods and a standardised benchmark for evaluating SD [9–12, 23]. Standards are still under development, including working by ISO [14], which will provide an overview of SD concepts, methods, uses, and considerations.

Data quality significantly impacts the performance of ML models, as “generative models are constrained by the size and quality of the training dataset” [6]. While quantitative metrics alone do not capture the full range of qualities desired in the output of generative models, human visual inspection can play a key role in the generative model validation workflow [3, 8, 18, 23]. Thus, a complete evaluation approach includes a “Human-in-the-loop” (HITL) or a “Doctor-in-the-loop” (DITL) [13] in the workflow. In it, domain knowledge is essential [23] to identify gaps when the model or the model developer may overlook complex relationships [19] or when a more well-rounded assessment of model performance is needed to catch artifacts [18].

Nevertheless, operationalizing the evaluation of SD quality with DITL presents several challenges, such as time constraints, as doctors have vital responsibilities within their practice and often may lack the availability to engage in a lengthy process to thoroughly assess large and high-dimensional synthetic datasets [20], while personal biases, fatigue, and human errors can introduce subjectivity and imprecision into assessments [7].

While recent studies have explored frameworks or protocols for qualitative expert assessment of synthetic medical data (SMD) targeting a specific data modality (i.e., tabular and image data [8, 23]), this study accommodates multiple data modalities and focuses on user interaction and approaches to user interface solutions for a DITL assessment, rather than defining data quality dimensions and related quantitative metrics. It is part of a larger research project that aims to develop a platform for healthcare ML and data engineers, doctors, and researchers to access trustworthy datasets, augmented with controlled data synthesis for experimentation and modelling purposes. The project’s use cases, representing specific medical research areas such as heart failure, Parkinson’s disease, diabetic retinopathy, epilepsy, and the associated electronic health records (EHRs), justify exploring multiple data modalities to assess the quality and suitability of such synthetic medical data for research.

We sought to provide the doctor/reviewer of the SMD with a guided and systematic review, not only to categorise the output, but also to provide information to improve the generative process.

This research contributes to the development of a DITL workflow for evaluating SMD and supports human-centered approaches to SMD assessment. Our paper includes a review of related work on the qualitative evaluation of SD and HITL-focused interfaces, an overview of the participatory design methods that shaped our preliminary designs, and the design sketches for the platform’s module dedicated to the qualitative evaluation of SMD. Additionally, we present key insights from domain experts on the challenges that SD technology poses to the healthcare ecosystem.

2 Related work

2.1 Evaluation of Synthetic Data

Quantitative evaluation of SD quality is usually performed by programmatically extracting quality metrics from data. These metrics can be grouped according to specific auditing goals (e.g., Improved Precision and Density to evaluate Fidelity, Improved Recall and Coverage for Diversity, Authenticity for Privacy, or Utility for Performance [1, 15, 17, 21]). Nonetheless, such metrics have limitations, including trade-offs between quality dimensions. For example, Fidelity may be sacrificed over Privacy, or Fairness may be sacrificed over Performance, and vice versa [15]. Modeling errors can have severe consequences for patients and providers, typically leading to risk aversion among healthcare stakeholders and skepticism and mistrust about using SD [9]. Regardless of the number of quality dimensions and metrics used to assess the quality of SD, they should be complemented by qualitative domain knowledge [23]. Qualitative data evaluation involves humans who may not have a deeper knowledge of ML systems, but do have deep domain expertise helpful in interpreting or evaluating model outputs and inputs, such as training data [7]. This expertise can contribute to data quality that is “fit for purpose” and representative of the real world [4], as “the conventional techniques used to generate synthetic data may inadequately capture the rich complexity and diverse array of real-world medical scenarios” [9].

Recent studies on SD generation in clinical contexts have often employed Visual Turing Tests [8] in which a clinician qualitatively evaluates image quality by determining whether an image is real or synthetic. Gatoula’s et al. proposed protocol for the Clinical Evaluation of Medical Image Synthesis by physicians [8], incorporates Visual Turing Tests as one of its components, alongside additional assessment procedures, addressing both quantitative and qualitative aspects. This protocol features questions to evaluate synthetic image quality, variety, plausibility concerning texture, anatomical structures, and diagnostic relevance. However, the proposed protocol focuses only on image data modality and includes real and synthetic images without reference to statistical metrics. Vallevik et al. present a comprehensive framework for quality assurance of synthetic data in healthcare [23]. It details the entire pipeline of synthetic data generation organised into up to six stages, each referring specific data quality considerations that should be assessed, but it lacks how to operationalise the assessment in terms of a user interaction flow.

2.2 Challenges in Designing DITL User Interfaces

Humans are not immune to fatigue, errors, or subjectivity in their judgment, and these factors can contribute to inconsistencies regarding human evaluation. Research suggests that careful interface design—especially the information and guidance provided—can improve the consistency of their feedback [7]. Another key element is determining the sample for review, which involves presenting model outputs for user evaluation [7]. This may include user manual selection or automated suggestions based on model properties. The main challenge is reducing user effort while collecting enough feedback to refine the model. Effective strategies focus on choosing

diverse and representative samples, avoiding redundancy [7]. Some authors have used cluster-based sampling [3, 5, 20] to help review, train, or label large volumes of data. Additionally, the interface should communicate the purpose of the ML process—for example, the model’s application, performance goals, or the insights users want to obtain [7]. This clarity supports task understanding, and careful design can further enhance feedback consistency. Finally, ML models inherently involve uncertainty, which can be difficult to convey in user interfaces (UI) [7]. Non-experts may perceive ML-based solutions as deterministic and flawless, making it challenging for them to understand the inherent variability, uncertainty, and potential for error of such systems. In healthcare, where concepts can be subjective or complex, communicating uncertainty is critical to helping users set realistic expectations and make informed decisions.

3 Methods

The aim of this qualitative study was to inform the design of a DITL workflow for the assessment of SMD and to contribute to improved human-centred approaches to SMD. This study was driven by insights gained from prior interactions with ML engineers responsible for creating the generative models, and doctors who might review the data synthetically generated by these models. These key stakeholders were invited to participate in collaborative workshops designed to capture and explore their perspectives on the evaluation of synthetic medical data. The workshops were conducted remotely using Microsoft Teams video conferencing software and Mural, a collaborative visual working platform.

3.1 Participants

Participants were recruited from a convenience sample of 9 ML engineers and 7 doctors. The ML engineers were part of the generative AI development team, working in specific data modalities—tabular, image, and time series—while the doctors represented different medical specialties, including geriatrics, cardiology, ophthalmology, and neurology. The workshops were facilitated by 2 design researchers.

3.2 Workshops

The team developed four workshops to discuss the proposed workflow and content for the DITL qualitative evaluation of SMD. Each session covered a specific medical use case and data: i) Workshop 1 focused on tabular data, such as the output of blood analysis, ii) Workshop 2 was dedicated to retinal fundus imaging in the context of diabetic retinopathy, iii) Workshop 3 focused on EEG¹ in the scope of epilepsy, and iv) Workshop 4 was on ECG² and heart conditions. Table 1 shows the number of participants and profiles per workshop.

The designs of the DITL evaluation workflow were first co-designed, and the workshops co-facilitated, with the ML engineers involved in the study to ensure that: the low-fidelity designs were also aligned with the ML engineers’ vision so they didn’t contain any gross misunderstandings of the metrics or visualizations; the proposed workshops were well-structured and effectively addressed

the key questions, while also resolving the doubts of the generative AI development team; and during the workshop session, the technical questions from the doctors about the AI models were answered in detail by the ML experts.

4 Early Design sketches for DITL evaluation workflow

This section describes the low-fidelity designs created for each session, as they had slightly different versions of the sketches since optimal representations of multidimensional data are often highly specific to the nature of the data [7]. However, all versions followed the same guided workflow, as shown in Figure 1. Building on the concept of parallel review processes commonly used in clinical practice (e.g., evaluation of clinical trials and procedures), we took inspiration from academic peer review platforms designed to assess the quality of contributions in scientific publications. Thus, the proposed process involved assigning evaluation categories and a self-assessment of reviewer confidence. The following describes each proposed step of the workflow discussed during the workshops:

4.1 Generation criteria

The reviewer starts by gaining a basic understanding of the criteria used to generate the SD and the training data³. This involves four main components (Appendix, Figure 5): information about the training data, the purpose of the generation (whose importance is highlighted in [9]), the generative model used, and generation parameters.

Key learnings were incorporated into this sketch. For instance, reviewers are cautioned that limitations in the real data can introduce biases in the SD, especially for underrepresented groups (Figure 1). Insights from ML engineers emphasise the importance of understanding the purpose behind the creation of SD. This understanding is crucial, as the interpretation of specific metrics and outputs may vary according to the goals of data generation. For instance, if the model’s objective is to improve image quality—such as sharpness or contrast—the synthetic outputs should reflect these specific attributes.

The reviewer also sees basic model details (e.g., capabilities and parameters), which might include configurations for targeting particular pathologies or diagnostic criteria (Figure 1).

4.2 Global analysis report

The global analysis report (Appendix, Figure 6) provides quantitative metrics that may help reviewers pinpoint potential issues in the subsequent visual and qualitative review. Users receive individual scores for Fidelity, Privacy, and Diversity, which assess the quality of SD based on statistical metrics (Figure 2). A global score summarising these dimensions offers a quick overview of overall data quality. To aid understanding of the data quality dimensions and related metrics, we provide a brief explanation and guidance for each computed quantitative metric that provides a score for a specific data quality dimension. For example, to explain Fidelity—how

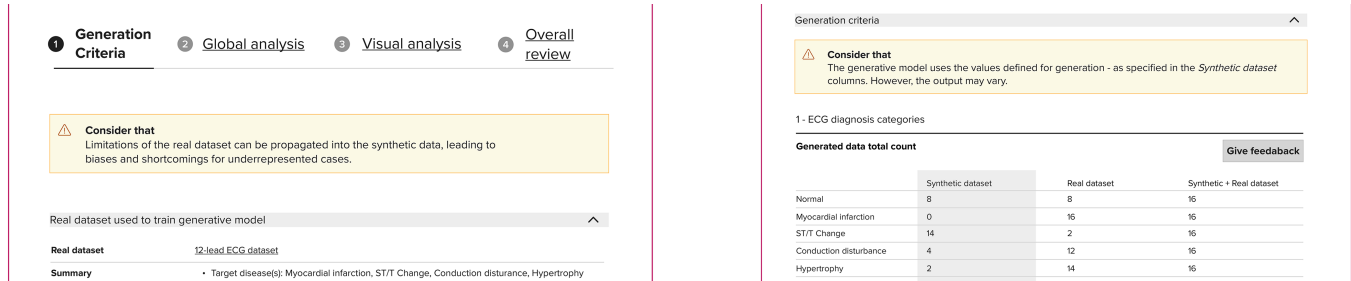
¹Intracranial Electroencephalography

²Electrocardiogram

³Training data here refers to the real data used to train the generative model

Table 1: Workshops to inform the user interface design to evaluate synthetic data.

Workshops	Number of participants and profiles	Scope
Workshop 1	2 doctors (1 geriatrician and 1 urologist/andrologist) and 2 ML engineers.	Qualitative evaluation workflow for tabular data.
Workshop 2	2 doctors (1 neurologist and 1 ophthalmologist) and 2 ML engineers.	Qualitative evaluation workflow for image – retinal fundus image.
Workshop 3	2 doctors (2 neurologists) and 3 ML engineers.	Qualitative evaluation workflow for time series – iEEG.
Workshop 4	2 doctors (1 geriatrician, and 1 cardiologist) and 2 ML engineers.	Qualitative evaluation workflow for time series – ECG.

**Figure 1: Sketch of tab element with the proposed DITL evaluation workflow (left) and details of SD generation criteria (right).**

similar SD is to real data—graphs and tables comparing the statistical properties of the real and SD were made available to provide an overview of the data generated versus the real data (Figure 2).

4.3 Visual analysis

Visual analysis (Appendix, Figure 7) is a key part of qualitatively evaluating SD, allowing reviewers to examine generated instances and provide feedback. Users can either randomly select instances for review or let the system automatically select them based on model properties, such as diagnostic categories and uncertainty levels (Figure 3). The goal of the latter is to direct doctors' feedback toward areas where the model struggles the most, enabling them to concentrate on the most impactful instances [2].

Each instance has basic information on its characteristics and buttons for acceptance or rejection based on quality. If rejected, users can provide additional feedback by selecting predefined issues that may be related to that data instance. Still, it is recommended to support a diverse range of input types [7], allowing doctors to express their feedback naturally. Thus, doctors may point out corrections, misclassification errors and provide free-form explanations for their assessments. For example, for a retinal fundus image, the user can select problems within the pathology classification, morphology or image quality (Figure 3).

4.4 Overall review

The overall review is the last step of this structured evaluation process (Appendix, Figure 8). In it, reviewers can offer their overall feedback regarding the dataset, outline their main concerns or problems, and be asked about their confidence in their own review (Figure 4).

Most doctors engaged in this study reported a lack of familiarity with SD. To account for this unfamiliarity, we suggested including a reviewer confidence assessment at the end of the workflow. An optional field for additional notes was also included in the sketch. Finally, reviewers would give their overall recommendation for the dataset by choosing an option such as Pass, Needs revision, Uncertain or Reject. With this feedback, ML engineers would know if the generated data needed to be refined, discarded or if the data passed the qualitative evaluation by clinical experts and appears to be well-suited for the target downstream task.

5 Results and discussion

5.1 Doctor-in-the-loop user interface workflow for synthetic medical data evaluation

A DITL approach to the qualitative assessment of SMD should guide doctors on what to evaluate, outline their responsibilities as reviewers, and include explanations for concepts they may be less familiar with.

In the Global analysis section of the workflow, among the statistical parameters and metrics presented, doctors felt more comfortable analysing 'realism', aka 'fidelity', of the SMD. Some of these metrics and graphs were commented on to make them more useful and in line with doctors' expectations. For example, in the context of an iEEG graph showing how well two signals (real and synthetic) were synchronised over time and across different frequencies, a neurologist noted that being able to interpret it was '*very dependent on which signals you choose. From the same patients or not, adjacent channels or not...*'. Participants also mentioned looking for inconsistencies in statistical values (mean, median, min-max) for healthy vs. pathological patterns, and fluctuations. Anatomical structures

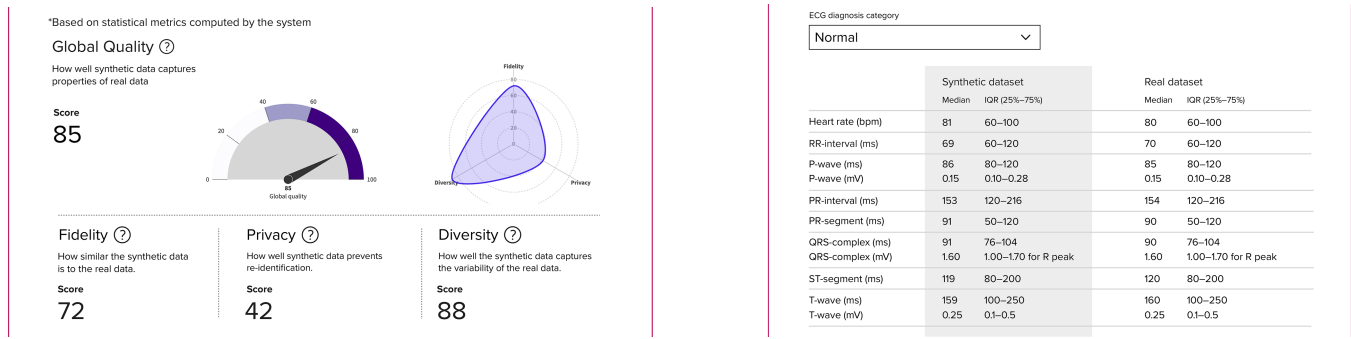


Figure 2: Sketch of quantitative metrics scores (left) and table comparing statistical properties between real and synthetic data (right).

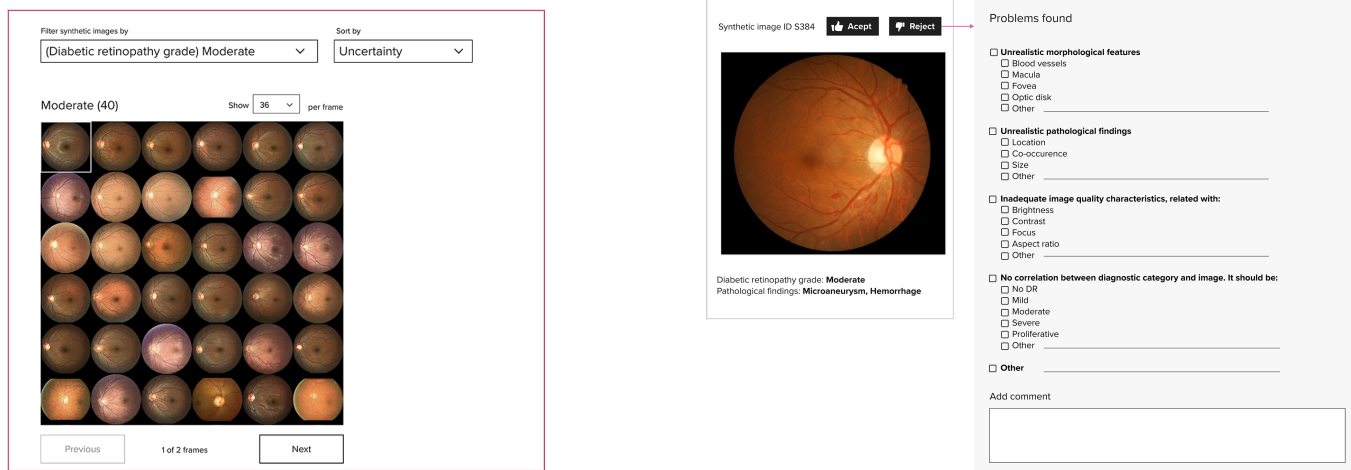


Figure 3: Sketch of the visual analysis (left) and form for entering problems found (right).

and specific features (e.g., location, size, color, brightness) were mentioned for retinal fundus images.

Although ML engineers often break down their development tasks into smaller units (e.g., generating a single synthetic iEEG channel), doctors found such outputs confusing because they typically analyse multiple channels together. Presenting synthetic data in a format that clinicians are used to is therefore crucial for gathering meaningful feedback.

In addition, it was explained that access to the diagnostic category of a synthetic instance or output would be very useful to frame its features in terms of what is morphologically and clinically possible. Participants had some difficulties with the data utility dimension, ‘(...) in clinical settings, the term “utility” can be misinterpreted as a concept of “clinical utility”’. It was suggested to clarify that the metric should be analysed from a ML perspective. In the light of participants’ questions and comments on the proposed data quality dimensions, Fidelity, Privacy, Diversity and Utility, and associated percentages, the interface may shift or hide this data, giving more prominence to the results and graphs of the statistical metrics that were found to be more useful.

In the visual analysis, the list of pre-defined problems with unrealistic features, incorrectly assigned diagnostic category, etc. was mostly aligned with the doctors’ vision. It was suggested that the instance should remain in view while the reviewer identifies the problems. It’s important to note that the image review sketch included a comparison between synthetic images and similar real instances from the training data. While some doctors have found this feature helpful, integrating it may pose challenges due to privacy concerns and the risk of real data leakage.

Finally, concerning the overall review, one of the participants suggested including a summary or synthesis of the issues identified by the reviewer while analysing the data, which seems sensible to facilitate the final review. These early drafts also featured a reviewer expertise form with predefined levels of expertise in analysing such data (e.g., “How experienced are you in iEEG analysis?”). Most participants disagreed with this approach, stating that there should be ways to ensure the assessment is carried out by a doctor or intern with expertise in the data modality from the very beginning, not a student or a doctor from an unrelated specialty.

The figure shows two side-by-side form sketches. The left form, titled 'Problems found', contains a list of checkboxes for issues: 'Unrealistic cardiac waveform, complex or segments morphologies', 'Unrealistic interval durations', 'Failed in the assigned diagnostic categories', 'Edge cases under-represented', 'Low diversity', and 'Other'. Below this is an 'Add comment' text area. The right form, titled 'About the reviewer', has two sections. The first, 'Reviewer expertise in medical field', asks 'How experienced are you in ECG analysis?' with radio button options: 'None (I am not familiar with this analysis)', 'Limited (I have basic knowledge in this analysis)', 'Moderate (I have some expertise in this analysis)', 'High (I have a strong background in this analysis)', and 'Expert (I am an expert in this analysis)'. The second section, 'Reviewer confidence', asks 'How confident are you in your review?' with radio button options: '1 = Not confident at all', '2 = Slightly confident', '3 = Somewhat confident', '4 = Quite confident', and '5 = Extremely confident'. Both sections have an 'Add comment' text area. A 'Submit' button is at the bottom right of the right form.

Figure 4: Sketch of the form with a summary of issues found (left) and a form for gathering reviewers' confidence (right).

5.2 Challenges for synthetic medical data and future work

5.2.1 Challenges through the lens of doctors. In general, doctors were unfamiliar with SMD and many of the concepts associated with the quantitative methods and metrics used by ML engineers to evaluate it. This lack of familiarity and scepticism towards SMD poses a significant challenge to their role as evaluators. Therefore, interfaces for SMD to be used by doctors should prioritise explainability to facilitate interpretation and increase clarity.

Like ML engineers, doctors are concerned about the fidelity of the data. They question whether it merely replicates real data or compromises privacy by being too similar to actual patient data. Another common challenge is the limited access to source data, which may fail to cover rare exceptions, as well as hidden underlying characteristics of medical data. These concerns underline the importance of a DITL assessment of SMD supported by a complete, transparent and explainable UI solution. The solution should also feature a robust feedback component to adequately capture qualitative feedback and inform model refinements.

As discussed during the study, it is important to define the purpose and use case for SMD to evaluate its characteristics accordingly. Once the synthetic data is generated and its quality validated as suitable for the target downstream task, doctors emphasised the importance of ensuring its responsible use for the intended purpose.

Finally, evaluating synthetic data can be laborious and tedious, even when employing sampling review strategies. Further interdisciplinary research is needed to assess how this process will fit into existing workflows of assessing SMD quality for clinical purposes.

5.2.2 Future work. Future work includes refining the SMD evaluation workflow design based on workshop feedback and future

developments on quality dimensions and quantitative metrics. This refinement will be followed by usability testing, implementation and evaluation of a pilot study to assess the prototype's user interface and workflow effectiveness and ease of use. It will also assess the fit and appropriateness with the users' existing processes and workflows. Nonetheless, this emerging topic lacks a one-size-fits-all solution, making participatory design approaches such as this one essential for developing a system that evaluates multimodal SMD responsibly.

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References

- [1] Ahmed Alaa, Boris Van Breugel, Evgeny S. Saveliev, and Mihaela van der Schaar. 2022. How Faithful is your Synthetic Data? Sample-level Metrics for Evaluating and Auditing Generative Models. In *Proceedings of the 39th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 162)*, Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvari, Gang Niu, and Sivan Sabato (Eds.). PMLR, Baltimore, Maryland, USA, 290–306. <https://proceedings.mlr.press/v162/aaa22a.html>
- [2] Saleema Amershi, Maya Cakmak, W. Bradley Knox, and Todd Kulesza. 2014. Power to the People: The Role of Humans in Interactive Machine Learning. *AI Magazine* 35, 4 (2014), 105–120. doi:10.1609/aimag.v35i4.2513

- [3] Yannick Assogba, Adam Pearce, and Madison Elliott. 2023. Large Scale Qualitative Evaluation of Generative Image Model Outputs. arXiv:2301.04518
- [4] Enrique Bernal-Delgado, Sarah Craig, Thomas Engsig-Karup, Francisco Estupiñán-Romero, Sahlertz Kristiansen Nina, and Bredmose Simonsen Jesper. 2022. *European Health Data Space Data Quality Framework*. Deliverable 6.1. TEHDAS - Towards the European Health Data Space. 40 pages. <https://tehdas.eu/results/>
- [5] Chelsea Chandler, Peter W Foltz, and Brita Elvevåg. 2022. Improving the Applicability of AI for Psychiatric Applications through Human-in-the-loop Methodologies. *Schizophrenia Bulletin* 48, 5 (Sept. 2022), 949–957. doi:10.1093/schbul/sbac038
- [6] Richard J. Chen, Ming Y. Lu, Tiffany Y. Chen, Drew F. K. Williamson, and Faisal Mahmood. 2021. Synthetic Data in Machine Learning for Medicine and Healthcare. *Nature Biomedical Engineering* 5, 6 (June 2021), 493–497. doi:10.1038/s41551-021-00751-8
- [7] John J. Dudley and Per Ola Kristensson. 2018. A Review of User Interface Design for Interactive Machine Learning. *ACM Transactions on Interactive Intelligent Systems* 8, 2 (June 2018), 1–37. doi:10.1145/3185517
- [8] Panagioti Gatoula, Dimitrios E. Diamantis, Anastasios Koulaouzidis, Cristina Carretero, Stefania Chetcuti-Zammit, Pablo Cortegoso Valdivia, Begoña González-Suárez, Alessandro Mussetto, John Plevris, Alexander Robertson, Bruno Rosa, Ervin Toth, and Dimitris K. Iakovidis. 2024. Clinical Evaluation of Medical Image Synthesis: A Case Study in Wireless Capsule Endoscopy. doi:10.48550/arXiv:2411.00178
- [9] Mauro Giuffrè and Dennis L. Shung. 2023. Harnessing the Power of Synthetic Data in Healthcare: Innovation, Application, and Privacy. *npj Digital Medicine* 6, 1 (Oct. 2023), 1–8. doi:10.1038/s41746-023-00927-3
- [10] Aldren Gonzales, Guruprabha Guruswamy, and Scott R. Smith. 2023. Synthetic Data in Health Care: A Narrative Review. *PLOS Digital Health* 2, 1 (Jan. 2023), e0000082. doi:10.1371/journal.pdig.0000082
- [11] Waldemar Hahn, Katharina Schütte, Kristian Schultz, Olaf Wolkenhauer, Martin Sedlmayr, Ulrich Schuler, Martin Eichler, Saptarshi Bej, and Markus Wolfien. 2022. Contribution of Synthetic Data Generation towards an Improved Patient Stratification in Palliative Care. *Journal of Personalized Medicine* 12, 8 (Aug. 2022), 1278. doi:10.3390/jpm12081278
- [12] Mikel Hernandez, Gorka Epelde, Ane Alberdi, Rodrigo Cilla, and Debbie Rankin. 2022. Synthetic Data Generation for Tabular Health Records: A Systematic Review. *Neurocomputing* 493 (July 2022), 28–45. doi:10.1016/j.neucom.2022.04.053
- [13] Andreas Holzinger. 2016. Interactive Machine Learning for Health Informatics: When Do We Need the Human-in-the-Loop? *Brain Informatics* 3, 2 (June 2016), 119–131. doi:10.1007/s40708-016-0042-6
- [14] International Organization for Standardization. 2025. ISO/IEC AWI TR 42103 – Information technology – Artificial intelligence – Overview of synthetic data in the context of AI systems (under development). ISO Standard (under development). <https://www.iso.org/standard/86899.html> Accessed: 2025-01-10.
- [15] James Jordon, Lukasz Szpruch, Florimond Houssiau, Mirko Bottarelli, Giovanni Cherubin, Carsten Maple, Samuel N. Cohen, and Adrian Weller. 2022. Synthetic Data – What, Why and How? doi:10.48550/arXiv.2205.03257 arXiv:2205.03257 [cs]
- [16] Jinah Kim, Sung-Ho Woo, Taekyung Kim, Won Tae Yoon, Jung Hwan Shin, Jee-Young Lee, and Jeh-Kwang Ryu. 2024. Development of a Cerebellar Ataxia Diagnosis Model Using Conditional GAN-based Synthetic Data Generation for Visuomotor Adaptation Task. *BMC Medical Informatics and Decision Making* 24, 1 (Nov. 2024), 336. doi:10.1186/s12911-024-02720-y
- [17] Tuomas Kynkäänniemi, Tero Karras, Samuli Laine, Jaakko Lehtinen, and Timo Aila. 2019. Improved Precision and Recall Metric for Assessing Generative Models. doi:10.48550/arXiv.1904.06991 arXiv:1904.06991 [stat]
- [18] John R. McNulty, Lee Kho, Alexandria L. Case, David Slater, Joshua M. Abzug, and Sybil A. Russell. 2024. Synthetic Medical Imaging Generation with Generative Adversarial Networks for Plain Radiographs. *Applied Sciences* 14, 15 (Jan. 2024), 6831. doi:10.3390/app14156831
- [19] Eduardo Mosqueira-Rey, Elena Hernández-Pereira, David Alonso-Ríos, José Bobes-Bascarán, and Ángel Fernández-Leal. 2023. Human-in-the-Loop Machine Learning: A State of the Art. *Artificial Intelligence Review* 56, 4 (April 2023), 3005–3054. doi:10.1007/s10462-022-10246-w
- [20] Eduardo Mosqueira-Rey, Alberto Pérez-Sánchez, Elena Hernández-Pereira, David Alonso-Ríos, José Bobes-Bascarán, Ángel Fernández-Leal, Vicente Moret-Bonillo, Yolanda Vidal-Ínsua, and Francisca Vázquez-Rivera. 2023. Human-in-the-Loop Machine Learning for the Treatment of Pancreatic Cancer. In *2023 International Joint Conference on Neural Networks (IJCNN)*. IEEE, Gold Coast, Australia, 1–9. doi:10.1109/IJCNN54540.2023.10191456
- [21] Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunje Choi, and Jaewon Yoo. 2020. Reliable Fidelity and Diversity Metrics for Generative Models. doi:10.48550/arXiv.2002.09797 arXiv:2002.09797 [cs]
- [22] Elena Sizikova, Andreu Badal, Jana G Delfino, Miguel Lago, Brandon Nelson, Niloufar Saharkhiz, Berkman Sahiner, Ghada Zamzmi, and Aldo Badano. 2024. Synthetic Data in Radiological Imaging: Current State and Future Outlook. *BJR/Artificial Intelligence* 1, 1 (March 2024), ubae007. doi:10.1093/bjrai/ubae007
- [23] Vibeke Binz Vallevik, Aleksandar Babic, Serena E. Marshall, Severin Elvatun, Helga M.B. Brøgger, Sharmini Alagaratnam, Bjørn Edwin, Narasimha R. Veer-aragavan, Anne Kjersti Befring, and Jan F. Nygård. 2024. Can I Trust My Fake Data – A Comprehensive Quality Assessment Framework for Synthetic Tabular Data in Healthcare. *International Journal of Medical Informatics* 185 (May 2024), 105413. doi:10.1016/j.ijmedinf.2024.105413
- [24] Bruno Vaz and Álvaro Figueira. 2025. GANs in the Panorama of Synthetic Data Generation Methods. *ACM Transactions on Multimedia Computing, Communications, and Applications* 21, 1 (Jan. 2025), 1–28. doi:10.1145/3657294
- [25] Adam Westerski and Wee Teck Fong. 2025. Synthetic Data for Object Detection with Neural Networks: State-of-the-Art Survey of Domain Randomisation Techniques. *ACM Transactions on Multimedia Computing, Communications, and Applications* 21, 1 (Jan. 2025), 1–20. doi:10.1145/3637064

A Appendices

ECG synthetic
For review

1 Generation Criteria
2 Global analysis
3 Visual analysis
4 Overall review

Consider that
Limitations of the real dataset can be propagated into the synthetic data, leading to biases and shortcomings for underrepresented cases.

Real dataset used to train generative model

Real dataset
12-lead ECG dataset

Summary

- Target diseases: Myocardial infarction, ST/T Change, Conduction disturbance, Hypertrophy
- Acquisition modality: ECG
- Standard set of 12 leads
- Recording time: 10 s
- Original frequency: 400 Hz
- Number of records (rows): 52
- Annotation classes: (Diagnosis categories) Normal, Myocardial infarction, ST/T Change, Conduction disturbance, Hypertrophy
- Human annotation by a cardiologist

Generation purpose

Balance train dataset class distributions to train predictive ML model

Real dataset
12-lead ECG diagnosis categories

Generative model used

Generative model name
Model name

Description
2 lines maximum

Generation options available
The generative model used provides control:

- 1 - ECG diagnosis categories
 - Normal
 - Myocardial infarction
 - ST/T Change
 - Conduction disturbance
 - Hypertrophy

Generation criteria

Consider that
The generative model uses the values defined for generation - as specified in the *Synthetic dataset* columns. However, the output may vary.

1 - ECG diagnosis categories

Generated data total count
Give feedback

	Synthetic dataset	Real dataset	Synthetic + Real dataset
Normal	8	8	16
Myocardial infarction	0	16	16
ST/T Change	14	2	16
Conduction disturbance	4	12	16
Hypertrophy	2	14	16

Figure 5: Sketch of the generation criteria user interface presented in the workshop for ECG time series.

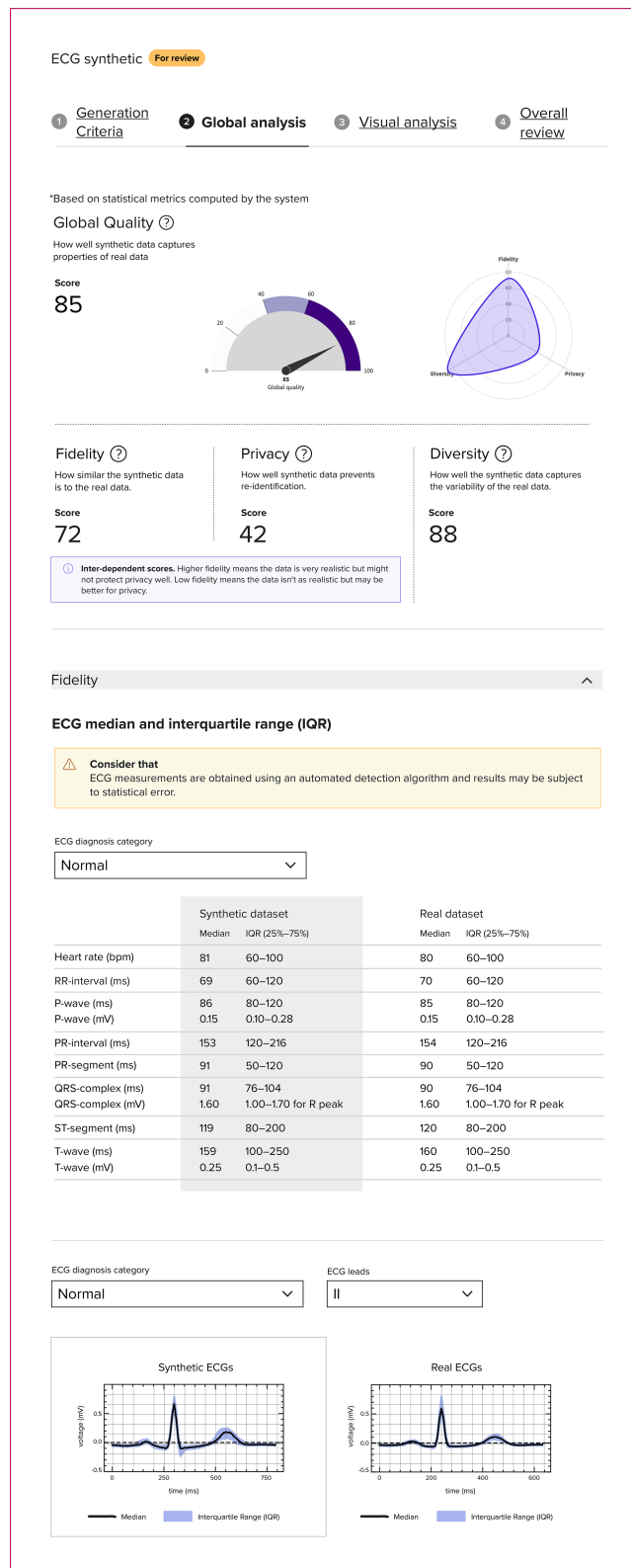


Figure 6: Sketch of the Global Analysis used in the workshops for ECG time series.

Synthetic retinal fundus images dataset For review

1 Generation Criteria

2 Global analysis

3 Visual analysis

4 Overall review

Synthetic set for inspection

195 out of 1950 retinal fundus images generated

Manual set selection ☐ OFF

1 Selection criteria

The synthetic set for review was randomly selected by the system to display 10% of the data generated per Diabetic retinopathy grade

Diagnostic category	Randomly selected	Total generated
No DR	0	0
Mild	10	100
Moderate	40	400
Severe	65	650
Proliferative	80	800
Total count	195	1950

Filter synthetic images by

(Diabetic retinopathy grade) Moderate

Sort by

Uncertainty

Moderate (40)

Show 36 per frame

Previous 1 of 2 frames Next

Synthetic image ID S384

Accept Reject

Diabetic retinopathy grade: Moderate

Pathological findings: Microaneurysm, Hemorrhage

Compare with closer real image

Real image ID R384

Reject synthetic image ID S384

Problems found

Unrealistic morphological features

☐ Blood vessels
 ☐ Macula
 ☐ Fovea
 ☐ Optic disk
 ☐ Other

Unrealistic pathological findings

☐ Location
 ☐ Co-occurrence
 ☐ Size
 ☐ Other

Inadequate image quality characteristics, related with:

☐ Brightness
 ☐ Contrast
 ☐ Focus
 ☐ Aspect ratio
 ☐ Other

No correlation between diagnostic category and image. It should be:

☐ No DR
 ☐ Mild
 ☐ Moderate
 ☐ Severe
 ☐ Proliferative
 ☐ Other

Other

Add comment

Submit review

Figure 7: Sketch of visual analysis of images. On the right is the visual analysis page. On the left is a sketch of a rejection form of a synthetic image instance.

The figure shows two wireframe sketches of a review interface. The left sketch is the 'Overall review' form, and the right sketch is the 'About the reviewer' form. A red arrow points from the 'Next' button in the left form to the 'About the reviewer' form.

Overall review form (Left):

- Header: ECG synthetic For review
- Progress bar: 1 Generation Criteria, 2 Global analysis, 3 Visual analysis, 4 Overall review
- Problems found**
 - ☐ Unrealistic cardiac waveform, complex or segments morphologies
 - ☐ Unrealistic interval durations
 - ☐ Failed in the assigned diagnostic categories
 - ☐ Edge cases under-represented
 - ☐ Low diversity
 - ☐ Other _____
- Add comment
- Reviewed set**
 - Select
 - Pass
 - Needs revision
 - Uncertain
 - Reject
- Add comment
- Next

About the reviewer form (Right):

- About the reviewer**
- Reviewer expertise in medical field**
 - How experienced are you in ECG analysis?
 - ☐ None (I am not familiar with this analysis)
 - ☐ Limited (I have basic knowledge in this analysis)
 - ☐ Moderate (I have some expertise in this analysis)
 - ☐ High (I have a strong background in this analysis)
 - ☐ Expert (I am an expert in this analysis)
- Add comment
- Reviewer confidence**
 - How confident are you in your review?
 - ☐ 1 - Not confident at all
 - ☐ 2 - Slightly confident
 - ☐ 3 - Somewhat confident
 - ☐ 4 - Quite confident
 - ☐ 5 - Extremely confident
- Add comment
- Submit

Figure 8: Sketch of the overall review on the left, and the reviewer confidence form on the right.